

Chapter 8.

Assignment problems.

Situation

In swimming competitions, a 4×100 metre medley relay involves four different swimmers, each swimming 100 metres and with each doing a different stroke.

The first swimmer swims 100 metres backstroke,
the second swimmer swims 100 metres breaststroke,
the third swimmer swims 100 metres butterfly,
and the fourth swimmer swims 100 metres freestyle.

A swimming club has four swimmers who will form the team for the men's 4×100 metres medley relay team. They are Alex, Ben, Chris and Devon. The season personal best times for each of these swimmers, in each of the strokes, over 100 metres are as given in the table below.

(Times are given as minutes.seconds.hundredths-of-a-second. Thus 1.08.72 means 1 minute, 8 seconds and 72 hundredths of a second. I.e. 68.72 seconds all together.)

	Backstroke	Breaststroke	Butterfly	Freestyle
Alex	1.08.72	1.20.17	56.28	58.17
Ben	1.01.24	1.09.21	1.10.21	53.05
Chris	1.03.21	1.17.16	1.03.27	55.21
Devon	1.04.17	1.08.71	1.17.24	54.37

Which swimmer would you have swimming each leg of the relay to give the fastest possible time for the team, based on the season personal best times?

Suppose instead that Alex's season personal best time for the butterfly is 1.08.17. I.e. the table of season personal best times would now be as follows:

	Backstroke	Breaststroke	Butterfly	Freestyle
Alex	1.08.72	1.20.17	1.08.17	58.17
Ben	1.01.24	1.09.21	1.10.21	53.05
Chris	1.03.21	1.17.16	1.03.27	55.21
Devon	1.04.17	1.08.71	1.17.24	54.37

What would be the preferred order now?

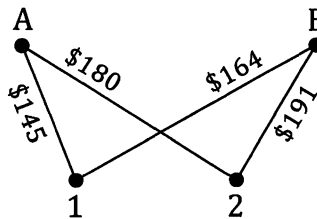
The situation on the previous page required us to assign each swimmer to a stroke in order to give the fastest medley relay team, based on season personal best times. This chapter considers **assignment problems** of this type, also called **allocation problems**. The swimming situation required us to allocate the swimmers in order to give the best total *time*. Not all problems of this type involve optimising time. Some may be concerned with optimising cost or distance travelled etc.

Assignment problems.

Suppose a company wishes to use two transport companies, A and B, for two deliveries, 1 and 2, giving one delivery to each transport company. The locations of each pick up point and delivery point in relation to each transport company, and the transport company costs, means that for each job, each of the transport companies quotes a different price. The quoted prices are as follows:

	Delivery 1	Delivery 2
Company A	\$145	\$180
Company B	\$164	\$191

Alternatively, this information could be given as a bipartite graph:



Can you see which allocation of the two jobs will minimise the total cost (remembering that we must give one delivery to each company)?

If we write AB to indicate company A doing delivery 1 and company B doing delivery 2, there are only two allocations that are possible:

AB and BA

AB: Company A doing delivery 1 and company B doing delivery 2.

$$\begin{aligned}\text{Total cost} &= \$145 + \$191 \\ &= \$336\end{aligned}$$

BA: Company B doing delivery 1 and company A doing delivery 2.

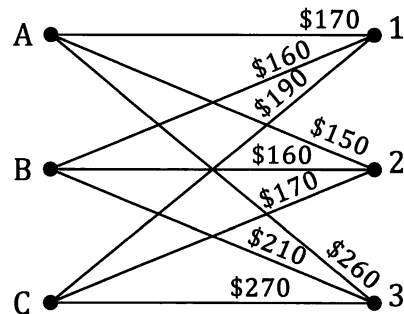
$$\begin{aligned}\text{Total cost} &= \$164 + \$180 \\ &= \$344\end{aligned}$$

Cheapest solution: Company A doing delivery 1 and company B doing delivery 2 for a total cost of \$336.

Is this the solution you chose just by considering the table of costs?

Now let us consider a situation involving three companies, A, B and C, and three deliveries, 1, 2 and 3, again with one delivery to be assigned to each company. Suppose the costs this time are as follows:

	1	2	3
A	\$170	\$150	\$260
B	\$160	\$160	\$210
C	\$190	\$170	\$270



Now, writing ABC to mean company A does delivery 1, B does delivery 2 and C does delivery 3, there are six possible ways the assignments can be organised:

ABC:	\$170	+	\$160	+	\$270	=	\$600	
ACB:	\$170	+	\$170	+	\$210	=	\$550	←
BAC:	\$160	+	\$150	+	\$270	=	\$580	
BCA:	\$160	+	\$170	+	\$260	=	\$590	
CAB:	\$190	+	\$150	+	\$210	=	\$550	←
CBA:	\$190	+	\$160	+	\$260	=	\$610	

Thus there are two allocations that equally give the minimum cost of \$550:

A do delivery 1, B do delivery 3 and C do delivery 2,
or
A do delivery 2, B do delivery 3 and C do delivery 1.

Allocating for a maximum.

Some allocation problems may involve situations in which the most desirable, or optimum, situation involves a maximum rather than a minimum. Consider for example allocating three workers to three machines. Each machine makes different components and each worker has different competencies on each machine. However the company wants to allocate each worker to a machine such that the total number of components produced is maximised. Suppose that the number of components produced in a day by each worker, A, B and C, on each machine, 1, 2 and 3, is as follows:

	Machine 1	Machine 2	Machine 3
Worker A	1020	950	1430
Worker B	760	920	1200
Worker C	950	1000	1450

Which worker should be assigned to each machine to maximise the total number of components produced by the three workers?

Try to find a solution before turning the page where a solution is presented. Perhaps set up a spreadsheet showing the various arrangements and their totals.

Writing ABC to mean worker A is on machine 1, B is on machine 2 and C is on machine 3, there are six possible ways the workers can be assigned to machines:

ABC:	1020	+	920	+	1450	=	3390	←	maximum total.
ACB:	1020	+	1000	+	1200	=	3220		
BAC:	760	+	950	+	1450	=	3160		
BCA:	760	+	1000	+	1430	=	3190		
CAB:	950	+	950	+	1200	=	3100		
CBA:	950	+	920	+	1430	=	3300		

Hence we should assign worker A to machine 1, worker B to machine 2 and worker C to machine 3 for a total number of components produced of 3390.

Four machines, four tasks.

The next example returns to an optimum situation involving a minimum but now a 4×4 table is involved.

Example 1

Four machines are to be assigned to carry out four tasks, with one machine for each task, and the aim of minimising the total machine time taken to complete the tasks.

The time each machine is estimated to complete each task is given below, in hours. Find the allocation of machines to tasks that will minimise total machine time.

	Task 1	Task 2	Task 3	Task 4
Machine A	15	18	9	12
Machine B	13	21	11	15
Machine C	12	19	8	13
Machine D	14	24	10	11

Writing ABCD to mean machine A is on task 1, machine B is on task 2, machine C is on task 3 and machine D is on task 4, there are 24 possible ways the machines can be assigned to the tasks:

ABCD: 15 + 21 + 8 + 11 = 55	CABD: 12 + 18 + 11 + 11 = 52
ABDC: 15 + 21 + 10 + 13 = 59	CADB: 12 + 18 + 10 + 15 = 55
ACBD: 15 + 19 + 11 + 11 = 56	CBAD: 12 + 21 + 9 + 11 = 53
ACDB: 15 + 19 + 10 + 15 = 59	CBDA: 12 + 21 + 10 + 12 = 55
ADBC: 15 + 24 + 11 + 13 = 63	CDAB: 12 + 24 + 9 + 15 = 60
ADCB: 15 + 24 + 8 + 15 = 62	CDBA: 12 + 24 + 11 + 12 = 59
BACD: 13 + 18 + 8 + 11 = 50 ←	DABC: 14 + 18 + 11 + 13 = 56
BADC: 13 + 18 + 10 + 13 = 54	DACB: 14 + 18 + 8 + 15 = 55
BCAD: 13 + 19 + 9 + 11 = 52	DBAC: 14 + 21 + 9 + 13 = 57
BCDA: 13 + 19 + 10 + 12 = 54	DBCA: 14 + 21 + 8 + 12 = 55
BDAC: 13 + 24 + 9 + 13 = 59	DCAB: 14 + 19 + 9 + 15 = 57
BDCA: 13 + 24 + 8 + 12 = 57	DCBA: 14 + 19 + 11 + 12 = 56

Assign machine A to task 2, B to task 1, C to task 3 and D to task 4. Total time 50 hours.

Are you able to set up a spreadsheet for the above listing that would be adaptable for any other "four machine four task" situation?

Exercise 8A

The following tables show the cost that trucking firms A, B, C, ... charge to deliver loads 1, 2, 3,

Assign trucking firms to loads such that in each case each firm is assigned a different load and the total cost is **minimised**.

1.

	Load 1	Load 2
Firm A	\$150	\$120
Firm B	\$120	\$110

2.

	Load 1	Load 2	Load 3
Firm A	\$100	\$140	\$150
Firm B	\$90	\$170	\$100
Firm C	\$110	\$140	\$130

3.

	Load 1	Load 2	Load 3	Load 4
Firm A	\$700	\$650	\$850	\$1200
Firm B	\$600	\$700	\$950	\$1050
Firm C	\$750	\$600	\$750	\$1250
Firm D	\$650	\$750	\$850	\$1150

The following tables show the number of components each operator, A, B, C, ... can produce in a day when operating machines 1, 2, 3,

Assign operators to tasks such that in each case each operator is assigned to a different machine and the total number of components produced in a day is **maximised**.

4.

	Machine 1	Machine 2
Operator A	80	90
Operator B	85	100

5.

	Machine 1	Machine 2	Machine 3
Operator A	210	200	170
Operator B	230	210	190
Operator C	200	180	180

6.

	Machine 1	Machine 2	Machine 3	Machine 4
Operator A	60	90	105	80
Operator B	80	90	110	85
Operator C	65	95	110	90
Operator D	75	85	115	75

An algorithm for solving allocation problems.

Whilst the process of listing all possibilities to determine the optimum arrangement is a valid process, there is an algorithm we can use to determine the optimum allocation, without having to list all possibilities. The steps of this algorithm are shown below.

Consider the situation of assigning four deliveries to four courier firms with each firm taking one delivery and with the allocation aiming to minimise the total cost.

Suppose the costs charged by each firm are as follows

	Delivery 1	Delivery 2	Delivery 3	Delivery 4
Courier firm A	\$250	\$100	\$140	\$120
Courier firm B	\$220	\$110	\$150	\$110
Courier firm C	\$220	\$90	\$160	\$130
Courier firm D	\$260	\$140	\$170	\$120

Step 1. Identify the **cost matrix**:

$$\begin{bmatrix} 250 & 100 & 140 & 120 \\ 220 & 110 & 150 & 110 \\ 220 & 90 & 160 & 130 \\ 260 & 140 & 170 & 120 \end{bmatrix}$$

Step 2. Subtract the smallest number in each row from every number in that row.

Hence we subtract 100 from each element of row 1,
110 from each element of row 2,
90 from each element of row 3,
and 120 from each element of row 4.

This gives the matrix:

$$\begin{bmatrix} 150 & 0 & 40 & 20 \\ 110 & 0 & 40 & 0 \\ 130 & 0 & 70 & 40 \\ 140 & 20 & 50 & 0 \end{bmatrix}$$

Step 3. Subtract the smallest number in each column from every entry in that column.

Hence we subtract 110 from each element of column 1,
0 from each element of column 2,
40 from each element of column 3,
and 0 from each element of column 4.

This gives the matrix:

$$\begin{bmatrix} 40 & 0 & 0 & 20 \\ 0 & 0 & 0 & 0 \\ 20 & 0 & 30 & 40 \\ 30 & 20 & 10 & 0 \end{bmatrix}$$

Step 4. Now draw as few **horizontal** and/or **vertical** straight lines as possible through all of the zeros. This question involves a 4×4 matrix so if the number of such lines equals 4, a solution can be found from this matrix, as is the case here.

$$\begin{bmatrix} 40 & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & 20 \\ \hline 0 & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & 0 \\ 20 & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & 30 & 40 \\ 30 & 20 & 10 & \begin{array}{|c|} \hline 0 \\ \hline \end{array} \end{bmatrix}$$

- Step 5. We now need to choose **one** zero in each column and **one** zero in each row.
 (Hint: First consider any rows and columns that only have one zero entry in them as that entry must be chosen.)

$$\begin{bmatrix} 40 & 0 & \boxed{0} & 20 \\ \boxed{0} & 0 & 0 & 0 \\ 20 & \boxed{0} & 30 & 40 \\ 30 & 20 & 10 & \boxed{0} \end{bmatrix}$$

The optimum situation is to assign courier firm A to delivery 3,
 courier firm B to delivery 1,
 courier firm C to delivery 2,
 and courier firm D to delivery 4.

This would have a total cost of \$140 + \$220 + \$90 + \$120 = \$570

- Note
- Step 4, the drawing of the lines could be done mentally.
 - If in step 4 the zeros are crossed through using less lines than there are rows (or columns) in the matrix, further adjustment of the matrix is necessary, as will be explained soon.
 - The listing of all 24 possible allocations of courier firms to deliveries is shown below. The reader should confirm that the optimum solution given by the algorithm is the same as obtained from the list of all possibilities.

ABCD:	\$250	+	\$110	+	\$160	+	\$120	=	\$640
ABDC:	\$250	+	\$110	+	\$170	+	\$130	=	\$660
ACBD:	\$250	+	\$90	+	\$150	+	\$120	=	\$610
ACDB:	\$250	+	\$90	+	\$170	+	\$110	=	\$620
ADBC:	\$250	+	\$140	+	\$150	+	\$130	=	\$670
ADCB:	\$250	+	\$140	+	\$160	+	\$110	=	\$660
BACD:	\$220	+	\$100	+	\$160	+	\$120	=	\$600
BADC:	\$220	+	\$100	+	\$170	+	\$130	=	\$620
BCAD:	\$220	+	\$90	+	\$140	+	\$120	=	\$570
BCDA:	\$220	+	\$90	+	\$170	+	\$120	=	\$600
BDAC:	\$220	+	\$140	+	\$140	+	\$130	=	\$630
BDCA:	\$220	+	\$140	+	\$160	+	\$120	=	\$640
CABD:	\$220	+	\$100	+	\$150	+	\$120	=	\$590
CADB:	\$220	+	\$100	+	\$170	+	\$110	=	\$600
CBAD:	\$220	+	\$110	+	\$140	+	\$120	=	\$590
CBDA:	\$220	+	\$110	+	\$170	+	\$120	=	\$620
CDAB:	\$220	+	\$140	+	\$140	+	\$110	=	\$610
CDBA:	\$220	+	\$140	+	\$150	+	\$120	=	\$630
DABC:	\$260	+	\$100	+	\$150	+	\$130	=	\$640
DACB:	\$260	+	\$100	+	\$160	+	\$110	=	\$630
DBAC:	\$260	+	\$110	+	\$140	+	\$130	=	\$640
DBCA:	\$260	+	\$110	+	\$160	+	\$120	=	\$650
DCAB:	\$260	+	\$90	+	\$140	+	\$110	=	\$600
DCBA:	\$260	+	\$90	+	\$150	+	\$120	=	\$620

Question 2 of Exercise 8A involved the following information:

	Load 1	Load 2	Load 3
Firm A	\$100	\$140	\$150
Firm B	\$90	\$170	\$100
Firm C	\$110	\$140	\$130

To minimise total cost the allocation was: Firm A to do load 1,
firm B to do load 3,
firm C to do load 2.
The total cost would then be \$340.

Follow the steps of the algorithm as shown below to see that it gives the same solution:

The cost matrix:

$$\begin{bmatrix} 100 & 140 & 150 \\ 90 & 170 & 100 \\ 110 & 140 & 130 \end{bmatrix}$$

Subtract the smallest number in each row from every number in that row.

$$\begin{bmatrix} 0 & 40 & 50 \\ 0 & 80 & 10 \\ 0 & 30 & 20 \end{bmatrix}$$

Subtract the smallest number in each column from every entry in that column.

$$\begin{bmatrix} 0 & 10 & 40 \\ 0 & 50 & 0 \\ 0 & 0 & 10 \end{bmatrix}$$

Draw as few horizontal and/or vertical straight lines as possible through all of the zeros. If the number of lines equals the number of rows (or columns) of the matrix then a solution can be found from this matrix.

$$\begin{bmatrix} \overset{\text{vertical line}}{\underset{\text{vertical line}}{0}} & 10 & 40 \\ \text{horizontal line} \overset{\text{vertical line}}{0} & 50 & \text{horizontal line} \overset{\text{vertical line}}{0} \\ \text{horizontal line} \overset{\text{vertical line}}{0} & 0 & \text{horizontal line} \overset{\text{vertical line}}{10} \end{bmatrix}$$

Determine the solution and relate identified locations in matrix back to the original.

$$\begin{bmatrix} \boxed{0} & 10 & 40 \\ 0 & 50 & \boxed{0} \\ 0 & \boxed{0} & 10 \end{bmatrix}$$

I.e., as before, firm A to do load 1 (\$100), firm B to do load 3 (\$100) and firm C to do load 2 (\$140). Total cost \$340.

If drawing the horizontal and/or vertical lines on the matrix to cover the zeros can be done using fewer lines than there are rows (or columns) in the matrix we need to adjust the matrix so that a solution can be found.

For example, consider:

	1	2	3
A	8	5	17
B	11	7	18
C	9	8	15

The cost matrix:

$$\begin{bmatrix} 8 & 5 & 17 \\ 11 & 7 & 18 \\ 9 & 8 & 15 \end{bmatrix}$$

Subtract the smallest number in each row from every number in that row.

$$\begin{bmatrix} 3 & 0 & 12 \\ 4 & 0 & 11 \\ 1 & 0 & 7 \end{bmatrix}$$

Subtract the smallest number in each column from every entry in that column.

$$\begin{bmatrix} 2 & 0 & 5 \\ 3 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

Draw as few horizontal and/or vertical straight lines as possible through all of the zeros. The number of lines is less than the number of rows (or columns) the matrix has. Further adjustments needed.

$$\begin{bmatrix} 2 & \textcircled{0} & 5 \\ 3 & \textcircled{0} & 4 \\ \textcircled{0} & \textcircled{0} & \textcircled{0} \end{bmatrix}$$

For the numbers that do not have a line drawn through them (2, 5, 3, 4), take the smallest of these numbers (2) from each of these numbers and also add this smallest number to any location where two of the straight lines intersect.

$$\begin{bmatrix} 0 & 0 & 3 \\ 1 & 0 & 2 \\ 0 & 2 & 0 \end{bmatrix}$$

The zeros now require three lines so a solution is possible (if not repeat the previous two steps). Determine the solution and relate the identified locations in matrix back to the original.

$$\begin{bmatrix} \boxed{0} & 0 & 3 \\ 1 & \boxed{0} & 2 \\ 0 & 2 & \boxed{0} \end{bmatrix}$$

A should be assigned to task 1. "Cost" is 8 units.

B should be assigned to task 2. "Cost" is 7 units.

C should be assigned to task 3. "Cost" is 15 units.

Total "cost" is 30 units.

The algorithm we are using here is called **The Hungarian Algorithm**, so named by Harold Kuhn in 1955, because the method was based on previous work by two Hungarian mathematicians, König and Egerváry.

Three examples of the algorithm are shown below. In each case the initial table shows the "cost" of allocating each of A, B, C, ... to tasks 1, 2, 3, ... This "cost" could be dollars, or hours, or distance travelled etc, and the task is to minimise the total cost. Check through each column to confirm that you too could obtain the solutions in this way.

	1	2	3	4
A	8	7	15	6
B	10	6	15	7
C	11	10	12	5
D	9	7	13	8

$$\begin{bmatrix} 8 & 7 & 15 & 6 \\ 10 & 6 & 15 & 7 \\ 11 & 10 & 12 & 5 \\ 9 & 7 & 13 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & 9 & 0 \\ 4 & 0 & 9 & 1 \\ 6 & 5 & 7 & 0 \\ 2 & 0 & 6 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cancel{0} & \cancel{1} & \cancel{3} & \cancel{0} \\ \cancel{2} & \cancel{0} & \cancel{3} & \cancel{1} \\ 4 & 5 & 1 & 0 \\ \cancel{0} & \cancel{0} & \cancel{0} & \cancel{1} \end{bmatrix}$$

$$\begin{bmatrix} \boxed{0} & 1 & 3 & 0 \\ 2 & \boxed{0} & 3 & 1 \\ 4 & 5 & 1 & \boxed{0} \\ 0 & 0 & \boxed{0} & 1 \end{bmatrix}$$

Assign A to 1. Cost = 8.
Assign B to 2. Cost = 6.
Assign C to 4. Cost = 5.
Assign D to 3. Cost = 13.
Total cost = 32.

	1	2	3	4
A	60	50	120	70
B	50	60	100	80
C	55	80	140	80
D	80	80	110	70

$$\begin{bmatrix} 60 & 50 & 120 & 70 \\ 50 & 60 & 100 & 80 \\ 55 & 80 & 140 & 80 \\ 80 & 80 & 110 & 70 \end{bmatrix}$$

$$\begin{bmatrix} 10 & 0 & 70 & 20 \\ 0 & 10 & 50 & 30 \\ 0 & 25 & 85 & 25 \\ 10 & 10 & 40 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 10 & 0 & 30 & 20 \\ 0 & 10 & 10 & 30 \\ 0 & 25 & 45 & 25 \\ \cancel{10} & \cancel{10} & \cancel{0} & \cancel{0} \end{bmatrix}$$

$$\begin{bmatrix} 10 & \boxed{0} & 20 & 10 \\ 0 & 10 & \boxed{0} & 20 \\ \boxed{0} & 25 & 35 & 15 \\ 20 & 20 & 0 & \boxed{0} \end{bmatrix}$$

Assign A to 2. Cost = 50.
Assign B to 3. Cost = 100.
Assign C to 1. Cost = 55.
Assign D to 4. Cost = 70.
Total cost = 275.

	1	2	3
A	80	60	90
B	100	70	110
C	90	100	100

$$\begin{bmatrix} 80 & 60 & 90 \\ 100 & 70 & 110 \\ 90 & 100 & 100 \end{bmatrix}$$

$$\begin{bmatrix} 20 & 0 & 30 \\ 30 & 0 & 40 \\ 0 & 10 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 20 & 0 & 20 \\ 30 & 0 & 30 \\ \cancel{0} & \cancel{10} & \cancel{0} \end{bmatrix}$$

$$\begin{bmatrix} \cancel{0} & \cancel{0} & \cancel{0} \\ 10 & \boxed{0} & 10 \\ \cancel{0} & \cancel{30} & \cancel{0} \end{bmatrix}$$

Multiple solutions.
Assign B to 2.
Cost = 70.
Then either
A to 1 (cost 80) and
C to 3 (cost 100).
Or
A to 3 (cost 90) and
C to 1 (cost 90).
Either way:
Total cost = 250

Suppose we want a maximum.

Question 6 of Exercise 8A involved assigning people to machines in such a way that the total number of components produced was **maximised**.

	Machine1	Machine 2	Machine 3	Machine 4
Operator A	60	90	105	80
Operator B	80	90	110	85
Operator C	65	95	110	90
Operator D	75	85	115	75

To use the Hungarian algorithm for a situation requiring a **maximum** we first note the largest number in the table, 115 in the table above, subtract every number in the table from this largest number, **and then proceed as we did when determining a minimum**.

Subtract each entry in the table from 115 to obtain the matrix on the right.

55	25	10	35
35	25	5	30
50	20	5	25
40	30	0	40

Subtract the smallest number in each row from every number in that row.

45	15	0	25
30	20	0	25
45	15	0	20
40	30	0	40

Subtract the smallest number in each column from every entry in that column.

15	0	0	5
0	5	0	5
15	0	0	0
10	15	0	20

Draw as few horizontal and/or vertical straight lines as possible through all of the zeros. If the number of lines equals the number of rows (or columns) of the matrix then a solution can be found from this matrix. If not, adjust as before.

15	0	0	5
0	5	0	5
15	0	0	0
10	15	0	20

Determine the solution and relate identified locations in matrix back to original.

15	0	0	5
0	5	0	5
15	0	0	0
10	15	0	20

I.e., assign operator A to machine 2, operator B to machine 1, operator C to machine 4 and operator D to machine 3.

Total number of components made is then 375 (= 90 + 80 + 90 + 115).

Check that this answer was the answer given for question 6 of exercise 8A.

Note: The Hungarian algorithm can be used when we are assigning one "person per task" and one "task per person". It relies on the fact, not proven here, that adding the same number to every element in a row or column of the cost matrix (or subtracting the same number from every element in a row or column of the cost matrix) does not change the "location" of the optimum arrangement. Once we have this location we can return to the original matrix to obtain the cost details.

Exercise 8B.

For each of questions 1 to 4 use the Hungarian algorithm to allocate the companies A, B, C, ... to the tasks 1, 2, 3, ... such that each company is allocated a different task and the total "cost" is **minimised**, where the units of "cost" of each item being assigned to each task is as given by the numbers in the table or bipartite graph. As well as stating the assignments of company to task, also state the total cost.

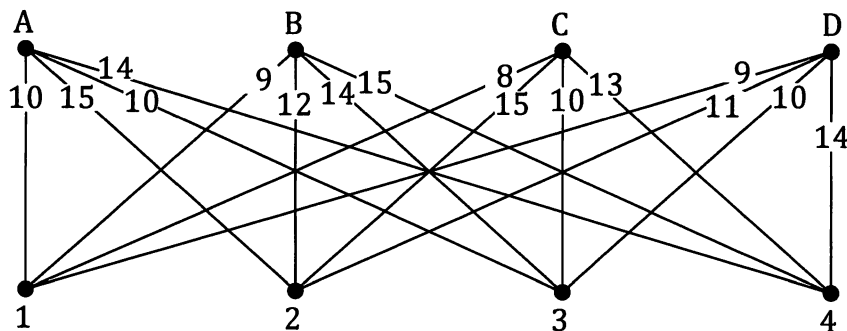
1.

	Task 1	Task 2
Company A	95	72
Company B	105	86

2.

	Task 1	Task 2	Task 3
Company A	70	110	120
Company B	50	140	150
Company C	100	170	140

3.



4.

	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6
Company A	20	30	20	30	20	30
Company B	24	20	12	35	10	28
Company C	30	25	20	55	21	38
Company D	25	44	21	36	20	30
Company E	20	32	20	40	22	35
Company F	40	50	20	48	25	40

For each of questions 5 to 9 use the Hungarian algorithm to allocate the companies A, B, C, ... to the tasks 1, 2, 3, ... such that each company is allocated a different task and the total "benefit" is **maximised**, where the units of "benefit" of each company being assigned to each task is as given by the numbers in the table or bipartite graph. As well as stating the assignment of company to tasks also state the benefit of each assignment.

5.

	Task 1	Task 2
Company A	250	120
Company B	240	90

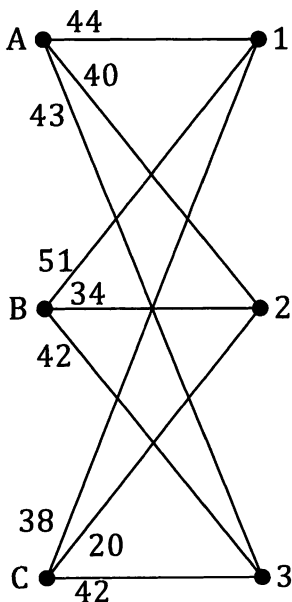
6.

	Task 1	Task 2	Task 3
Company A	29	29	31
Company B	26	24	28
Company C	22	33	22

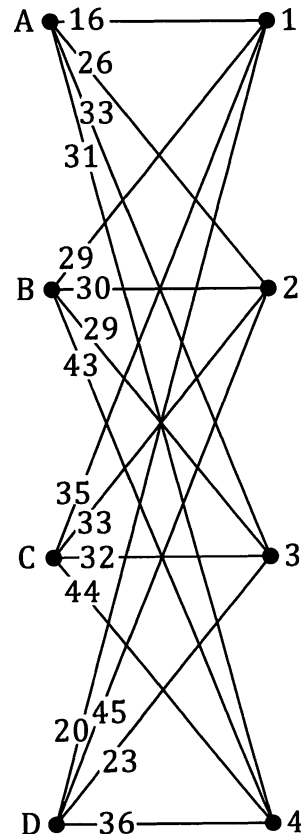
7.

	Task 1	Task 2	Task 3	Task 4
Company A	120	240	120	560
Company B	150	250	140	620
Company C	110	220	100	610
Company D	160	210	90	540

8.



9.



10. A company has four regional "sales reps" with one situated in Adelaide, one in Brisbane, one in Hobart and one in Perth. The company plans to open offices in Broome, Darwin, Melbourne and Sydney and, at least initially, wants each of the existing sales reps to oversee the new reps in these new offices, with one existing rep per new office. This overseeing role is likely to involve a fair bit of travel so the company decides to attach each existing rep to the new office in a way that minimises the total cost of a return airfare for each existing rep to visit the new office they are overseeing.

The return airfares between the locations are as follows:

	Broome	Darwin	Melbourne	Sydney
Adelaide	\$1080	\$780	\$240	\$380
Brisbane	\$1460	\$570	\$340	\$270
Hobart	\$1750	\$1120	\$290	\$550
Perth	\$720	\$680	\$440	\$580

- (a) Just by looking at the table try to write down what you think the optimal allocation of existing reps to new offices would be.
- (b) Clearly showing your use of the Hungarian algorithm determine the optimal allocation.
11. (a) Allocate four delivery jobs to four courier companies with each company getting one of the jobs and total costs being kept to a minimum, and state this minimum, given that the cost of each job with each company is as shown below.

	Job 1	Job 2	Job 3	Job 4
Quick Courier Co.	\$290	\$70	\$290	\$150
Speedy Courier Co.	\$320	\$70	\$260	\$160
Deliverit Couriers	\$460	\$80	\$320	\$180
Competitive Couriers	\$280	\$60	\$250	\$140

- (b) Suppose the job description for job 4 changed, causing every one of the four courier companies to increase their price for job 4 by \$50, all other jobs remaining the same as in the above table. Would this alter the optimum allocation of company to job? If it would, state the new allocation.
- (c) Suppose instead that it was job 2 that had its job description changed and this caused all four companies to multiply their price for that job by 4, all other jobs remaining as in the table. Would this alter the optimum allocation of company to job? If it would, state the new allocation.
- (d) Suppose instead that just one of the companies is to be used for all four jobs. Which company would this be if again the aim is to minimise total cost?

12. A real estate company is allocating areas to each of its five salespeople, Jack, Jim, Judy, Nicci and Ti. Each of these five salespeople is to have their own area, with the four areas labelled 1, 2, 3, 4 and 5.

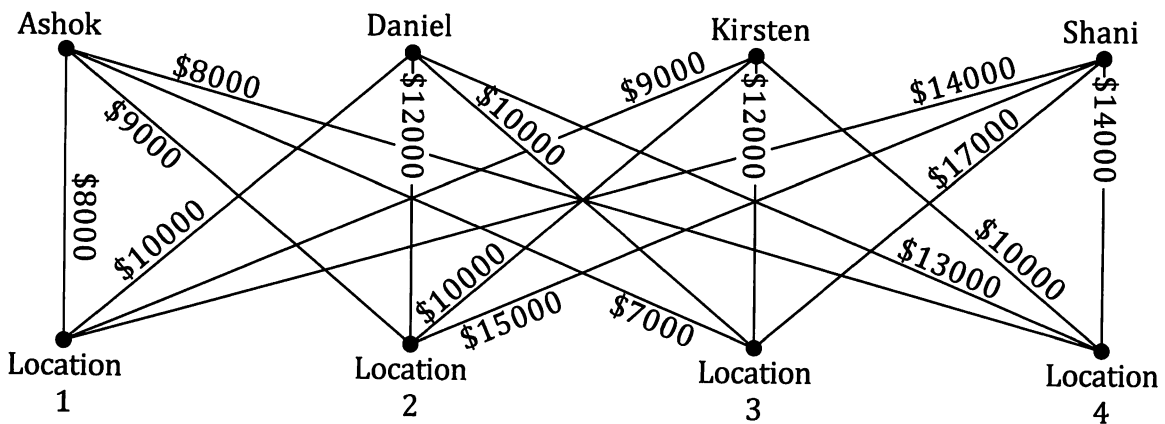
An analysis of the areas, the selling history of the representatives and their existing profile in the areas suggest that the number of houses each representative should sell in a year, were they assigned to a particular area, would be as follows.



	Area 1	Area 2	Area 3	Area 4	Area 5
Jack	10	9	8	7	6
Jim	5	7	7	9	7
Judy	9	7	9	10	8
Nicci	8	6	10	10	10
Ti	11	12	8	13	10

Allocate the representatives to the areas in such a way that will maximise the total number of houses the five representatives should sell in a year.
What will this maximum total be?

13. A mining company employs four new engineers, Ashok, Daniel, Kirsten and Shani, and wishes to locate one at each of its four major mining locations. Travel and relocation expenses are available to all four and the company decides to send the engineers to the location that minimises the total travel and relocation costs. These costs are estimated to be as follows.



Allocate each engineer to the appropriate mining location.

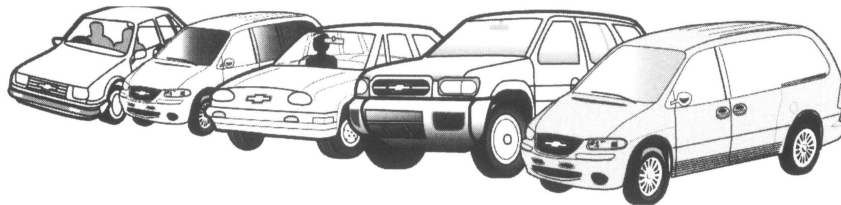
14. A construction company has one mobile crane available from each of its five equipment storage bases. It needs to send these five mobile cranes to five job locations, with one crane per location. The distances involved for each crane to get to each location are as shown in the following table, in kilometres.

	Location 1	Location 2	Location 3	Location 4	Location 5
Crane A	40	90	30	70	60
Crane B	30	70	40	60	30
Crane C	40	30	70	40	30
Crane D	50	80	40	50	90
Crane E	50	60	40	40	70

Allocate each crane to a location such that the total distance travelled is minimised.

15. A car making company has five factories, A, B, C, D and E each producing all of the five major models of car that the company produces:

The Arcain. The Bijou. The Charger. The Devine. The Exeon.



The company wishes to reorganise the factories and have one factory only producing the Arcain, another only producing the Bijou and so on.

Estimates suggest that the numbers each factory could produce of each model in a year, if they were dedicated to only producing the one model, would be as follows.

	Arcain	Bijou	Charger	Devine	Exeon
Factory A	230000	250000	200000	190000	180000
Factory B	150000	150000	130000	120000	120000
Factory C	320000	330000	300000	280000	290000
Factory D	180000	170000	120000	150000	140000
Factory E	240000	200000	200000	180000	180000

The Bijou is the best selling model so the company knows it wants to locate the construction of that model at its biggest factory, Factory C.

Allocate a factory to each model so each factory produces only one model, all five models are produced, the Bijou is at factory C, and the total number of models produced in a year is maximised.

Suppose the number of people $\neq$ the number of tasks.

In all of the assignment situations encountered so far in this chapter the number of items we have been assigning (be they swimmers, trucking companies, machine operators, etc) has always been the same as the number of things we have assigned these items to (be they swimming legs in a relay, loads to be delivered, machines, etc.). Such assignment problems are sometimes referred to as being *balanced*. The cost matrix will be a square matrix. Suppose the number of items we are assigning does not match the number of items we are assigning them to? In such cases we assign a *dummy row*, or *dummy column*, as appropriate, with all of its entries equal to zero. In this way we **create a square matrix** and the Hungarian algorithm can be used. The next two examples illustrate this technique.

Example 2

The director of a company decides that he wants a representative from the company at each of the three international trade conferences that are coming up. The director wishes to send a different manager to each, chosen from the four managers the company has based at its various branches, making the selection on the basis of the arrangement that keeps travel costs to a minimum. The associated travel costs are as follows:

	Conference 1	Conference 2	Conference 3
Manager A	\$1040	\$780	\$2140
Manager B	\$1150	\$560	\$1750
Manager C	\$1780	\$1250	\$1350
Manager D	\$1560	\$790	\$1215

What should the allocation of managers be?

Inserting a dummy column with all entries equal to zero gives the square matrix on the right.

1040	780	2140	0
1150	560	1750	0
1780	1250	1350	0
1560	790	1215	0

With a zero in each row subtracting the lowest number in each row from every entry in that row will leave the matrix unchanged.

Subtract the smallest number in each column from every number in that column.

0	220	925	0
110	0	535	0
740	690	135	0
520	230	0	0

Four horizontal and/or vertical lines are needed to cover all of the zeros so a solution can be found from this matrix.

0	220	925	0
110	0	535	0
740	690	135	0
520	230	0	0

Determine the solution:

$$\begin{bmatrix} 0 & 220 & 925 & 0 \\ 110 & 0 & 535 & 0 \\ 740 & 690 & 135 & 0 \\ 520 & 230 & 0 & 0 \end{bmatrix}$$

Relate identified locations in the matrix back to original, remembering that the final column is for a non existent conference.

$$\begin{array}{c} \begin{matrix} & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \end{matrix} \begin{bmatrix} 1040 & 780 & 2140 & 0 \\ 1150 & 560 & 1750 & 0 \\ 1780 & 1250 & 1350 & 0 \\ 1560 & 790 & 1215 & 0 \end{bmatrix} \end{array}$$

Hence send manager A to conference 1. (For a cost of \$1040.)
 Send manager B to conference 2. (For a cost of \$560.)
 Send manager D to conference 3. (For a cost of \$1215.)
 (Total cost \$2815.)



Instead of making a dummy column with all entries equal to zero try the previous example with the dummy column having all of its entries equal to the largest number in the matrix (i.e. 2140). Does that give the same solution?



Example 3

An international company has three locations across the world, each with its own director. The three directors decide they should each make a fact finding trip to one of four other countries to explore export opportunities. They choose four countries and decide that between them they will visit three of these four with each director visiting one country. The associated travel costs for each director to visit each country are as follows:

	Country 1	Country 2	Country 3	Country 4
Director A	\$10100	\$13300	\$11500	\$7200
Director B	\$9300	\$8200	\$7900	\$6300
Director C	\$12700	\$11700	\$14200	\$9500

Allocate each director to one of the countries so that each is attending a different country from the other two and total travel costs are kept to a minimum.

Inserting a dummy row with all entries equal to zero gives the square matrix on the right.

$$\begin{bmatrix} 10100 & 13300 & 11500 & 7200 \\ 9300 & 8200 & 7900 & 6300 \\ 12700 & 11700 & 14200 & 9500 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Subtracting the smallest number in each row from every number in that row gives the matrix on the right.

$$\begin{bmatrix} 2900 & 6100 & 4300 & 0 \\ 3000 & 1900 & 1600 & 0 \\ 3200 & 2200 & 4700 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

With a zero in each column, subtracting the smallest number in each column will not change the matrix.

Drawing horizontal and vertical lines indicates further adjustment needed.

$$\begin{bmatrix} 2900 & 6100 & 4300 & 0 \\ 3000 & 1900 & 1600 & 0 \\ 3200 & 2200 & 4700 & 0 \\ \hline 0 & 0 & 0 & 0 \end{bmatrix}$$

Adjust:

$$\begin{bmatrix} 1300 & 4500 & 2700 & 0 \\ 1400 & 300 & 0 & 0 \\ 1600 & 600 & 3100 & 0 \\ 0 & 0 & 0 & 1600 \end{bmatrix}$$

Drawing lines indicates yet further adjustment needed.

$$\begin{bmatrix} 1300 & 4500 & 2700 & 0 \\ \hline 1400 & 300 & 0 & 0 \\ 1600 & 600 & 3100 & 0 \\ \hline 0 & 0 & 0 & 1600 \end{bmatrix}$$

Adjust:

$$\begin{bmatrix} 700 & 3900 & 2100 & 0 \\ 1400 & 300 & 0 & 600 \\ 1000 & 0 & 2500 & 0 \\ 0 & 0 & 0 & 2200 \end{bmatrix}$$

Drawing lines indicates a solution can now be determined.

$$\begin{bmatrix} 700 & 3900 & 2100 & 0 \\ 1400 & 300 & 0 & 600 \\ 1000 & 0 & 2500 & 0 \\ \hline 0 & 0 & 0 & 2200 \end{bmatrix}$$

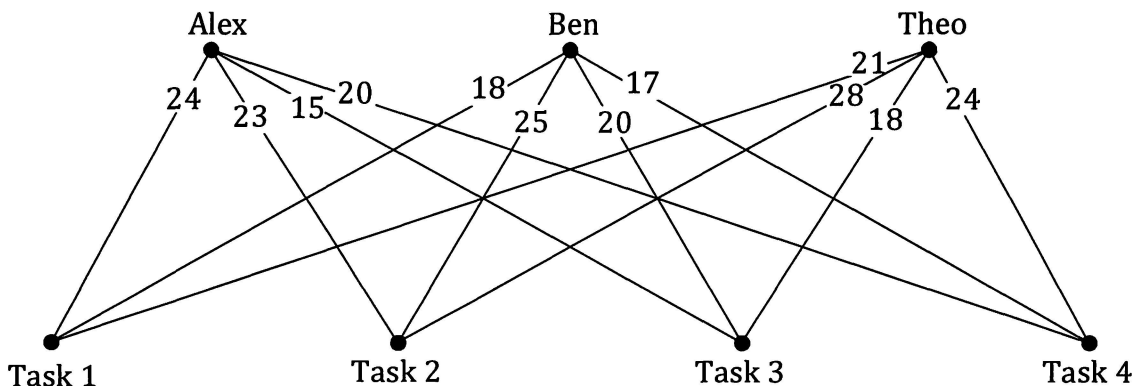
Determine the solution and relate identified locations in matrix back to original, remembering that the bottom row is for a non existent director.

$$\begin{bmatrix} 700 & 3900 & 2100 & \boxed{0} \\ 1400 & 300 & \boxed{0} & 600 \\ 1000 & \boxed{0} & 2500 & 0 \\ \boxed{0} & 0 & 0 & 2200 \end{bmatrix}$$

Director A should visit country 4. (Cost \$7200.)
 Director B should visit country 3. (Cost \$7900.)
 Director C should visit country 2. (Cost \$11700.)
 (Total cost \$26800.)

Exercise 8C.

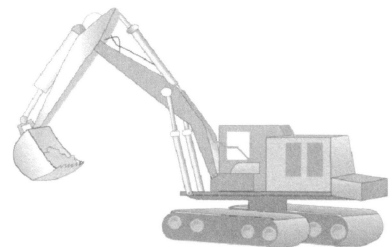
1. Three people, Alex, Ben and Theo, are each to be assigned a task from tasks 1, 2, 3 and 4, with each person having just one task, i.e. just three of the tasks will be assigned. The time each person takes to do each task, in minutes, is as shown below. The assignment of each person to a task is to be done in such a way that the total of the three times is to be minimised.



Under this minimal total time arrangement, if all three people start their assigned task at the same time, how many minutes later will all three have finished (assume any early finishers do not assist others with their task)? Is there a different allocation arrangement that would reduce this "time to finish three jobs when working simultaneously"?

2. An equipment hire company is asked to supply three industrial diggers, one to a worksite in Denton, one to a worksite in Abbysix and one to a worksite in Crampon.

The company has four such diggers available, one at the Horton storage facility, one at the Peel storage facility, one at the Washton storage facility and the fourth at the Wimsley storage facility. The distances, in kilometres, that each of these storage facilities is from each of the worksites is given in the following table:



	Denton	Abbysix	Crampon
Horton storage facility	27 km	25 km	25 km
Peel storage facility	58 km	22 km	18 km
Washton storage facility	34 km	36 km	51 km
Wimsley storage facility	31 km	55 km	54 km

Clearly showing your use of the Hungarian algorithm, assign three of the diggers, one to each worksite, such that the total distance to be travelled by the diggers to the worksites is minimised, and state this minimum distance.



3. A taxi company has four available taxis at various locations. The company receives three calls from separate individuals, Bill, Zarani and Tienielle, each requesting a taxi.



The estimated times each available taxi will take to get to each of the people requesting a taxi are as shown in the table below.

	Bill	Zarani	Tienielle
Taxi A	9	7	8
Taxi B	11	7	10
Taxi C	10	6	8
Taxi D	8	8	9

Assign a taxi to each person such that the total time taken by the three taxis used is the minimum possible, and state the time each person has to wait.

4. Four people from A, B, C, D and E are to be assigned jobs 1, 2, 3 and 4 with a different person for each job and one of the five people "missing out". The profit that will be made from each person doing each job is as shown in the table below.

	Job 1	Job 2	Job 3	Job 4
A	\$75	\$175	\$150	\$120
B	\$90	\$220	\$150	\$130
C	\$120	\$240	\$170	\$160
D	\$100	\$200	\$160	\$140
E	\$90	\$180	\$150	\$150

Supposing you are going to use the Hungarian algorithm to assign people to jobs according to the above requirements and in a way that maximises the profit to the company.

Which are you going to do first –

put in the dummy column and then subtract each number from the largest number,
or

subtract each number from the largest number and then put in the dummy column,
or doesn't it matter? Investigate, and find the solution to the assignment problem too, stating which person gets which job and the maximum profit.

Challenge.

Time and motion experts are looking at the assignment of ten workers to ten tasks, one worker per task. The experts first measure the time taken by each of the workers to complete each task. They find that the varied experience and skill levels of the workers mean that not all workers complete the same task in the same time. The various times taken by each worker on each task, in minutes, is as given below.

	Tasks									
	1	2	3	4	5	6	7	8	9	10
Worker A	17	21	9	28	15	19	33	45	15	14
Worker B	15	19	10	25	17	15	28	37	28	19
Worker C	22	22	13	21	17	22	47	46	17	17
Worker D	16	22	10	23	16	17	29	39	22	21
Worker E	18	20	9	29	21	18	32	40	23	14
Worker F	18	21	11	30	17	24	33	38	19	15
Worker G	17	23	12	22	16	18	45	40	24	21
Worker H	25	20	15	23	15	28	29	42	25	22
Worker I	16	20	10	25	14	17	32	55	20	18
Worker J	17	24	12	28	17	21	28	36	20	15

At present the workers are assigned as follows:

Worker A to task 1	(17 Mins)	Worker B to task 2	(19 Mins)
Worker C to task 3	(13 Mins)	Worker D to task 4	(23 Mins)
Worker E to task 5	(21 Mins)	Worker F to task 6	(24 Mins)
Worker G to task 7	(45 Mins)	Worker H to task 8	(42 Mins)
Worker I to task 9	(20 Mins)	Worker J to task 10	(15 Mins)

Summing these times gives a total of 239 minutes.

The experts want to find the allocation of worker to task that would minimise this total.

1. Solve the above allocation problem yourself using the Hungarian algorithm.
2. See if there is an online Hungarian algorithm solver and, if so, see if it gives the same solution to the above problem as you obtained.

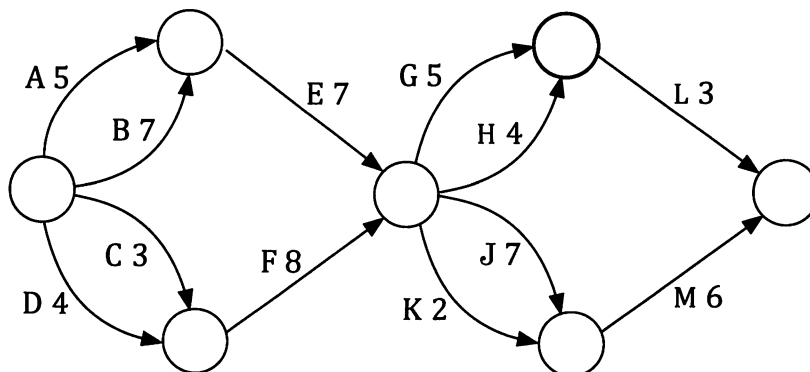
Miscellaneous Exercise Eight.

This miscellaneous exercise may include questions involving the work of this chapter, the work of any previous chapters, and the ideas mentioned in the preliminary work section at the beginning of the book.

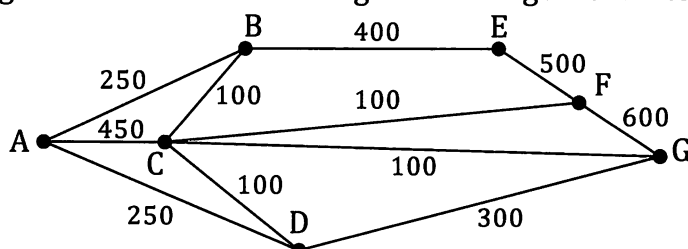
1. Copy and complete the following table. (1 yr = 12 months = 52 weeks = 365 days.)

Compounding frequency.	Nominal annual interest rate (%).	Effective annual interest rate (%).
Annual	10%	
Six monthly	10%	
Quarterly	10%	
Monthly	10%	
Weekly	10%	
Daily	10%	

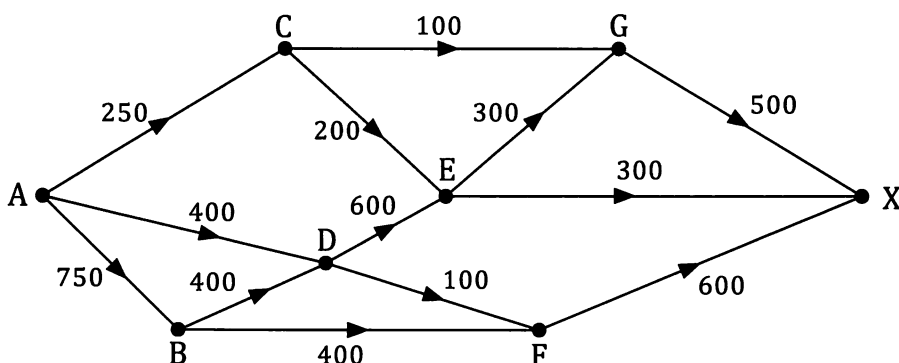
2. To the nearest dollar, how much should be invested in an annuity paying 10% annual interest, compounded annually, to provide a regular annual income of \$65000 for exactly 20 years?
Determine the answer by
(a) using a recursive formula with an initial "guess" of \$560 000,
and
(b) using the financial capability of some calculators or computer programs.
3. Suppose instead that the annual interest rate in the previous question had been 7.5% rather than 10%. Initially make a guess at how much would need to be invested to provide the same annual income, i.e. \$65000 for exactly 20 years, and then use recursion to determine the correct answer, to the nearest dollar.
4. A loan for \$8000 is taken out with compound interest charged at a rate of 8% per annum compounded monthly. If repayments of \$250 are made at intervals of one month after the start of the loan how much is owed immediately after the fifteenth repayment has been made?
5. The times for tasks A, B, C etc., shown in the project network below are in hours. (The letter I has been deliberately omitted.)
Determine the minimum completion time and critical path for the network.



6. The network below shows the connections in a telecommunications system. The vertices represent the switching stations, capable of sending messages from one station to another. The numbers on the edges give the number of messages that can be sent from one station to another in an hour.
- (a) Find the greatest number of messages that can go from A to G in an hour.
- (b) Find the greatest number of messages that can go from A to D in an hour.



7. The network below shows a system of pathways that allow units to flow from a source A to a sink X. The units that can flow along each pathway are as indicated by the numbers next to each pathway. Find the maximum number of units that can flow from A to X (and check your answer by finding a cut equal to your maximum flow).



If the capacity of pathway EX were to be increased by 300 units how would this alter the maximum flow from A to X?

8. Four swimmers, Julie, Micah, Fran and Marta are chosen to swim the medley relay for their school at the interschool swimming carnival. Each of the girls will swim one "leg" of the relay. One will swim 50 metres backstroke, one will swim 50 metres breaststroke, one will swim 50 metres butterfly and one will swim 50 metres freestyle.

To decide which girl swims each leg the coach times each of them over 50 metres of each style. The times recorded, in seconds, were as follows:

	Backstroke	Breaststroke	Butterfly	Freestyle
Julie	48.6	56.8	43.6	38.4
Micah	56.8	61.3	55.8	42.4
Fran	54.2	52.4	62.7	64.7
Marta	72.4	48.2	65.3	62.3

Use these times to suggest the most appropriate allocation of girls to strokes and suggest the total time for such a team.

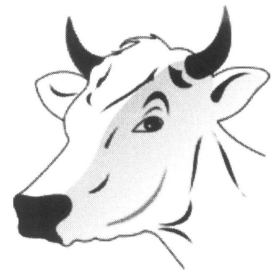
9. The following table gives the numbers of live cattle exported from Australia each year from 1988 to 2013. (Numbers are given to the nearest one hundred).

Year	1988	1989	1990	1991	1992	1993
Number of cattle	81500	83300	90200	94700	132300	190400
Year	1994	1995	1996	1997	1998	1999
Number of cattle	290900	480800	724100	912900	597000	833700
Year	2000	2001	2002	2003	2004	2005
Number of cattle	887000	797900	955100	689400	570800	531200
Year	2006	2007	2008	2009	2010	2011
Number of cattle	609200	668900	815600	921300	821600	621900
Year	2012	2013				
Number of cattle	516500	726000				

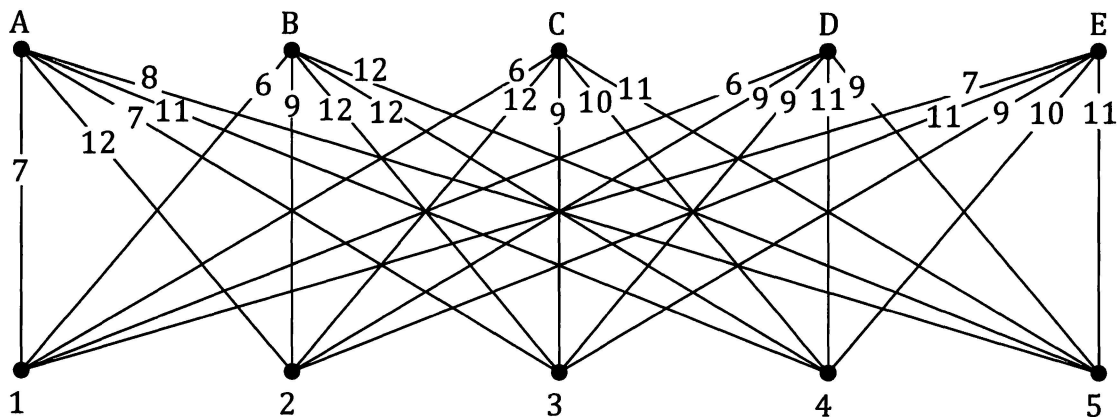
Source of data: Australian Bureau of Statistics.

With the assistance of a computer spreadsheet draw both the raw data and the five point moving averages on the same time series line graph.

Write some sentences describing any trends in the numbers of live cattle exported from Australia per year over the period of time from 1988 to 2013.



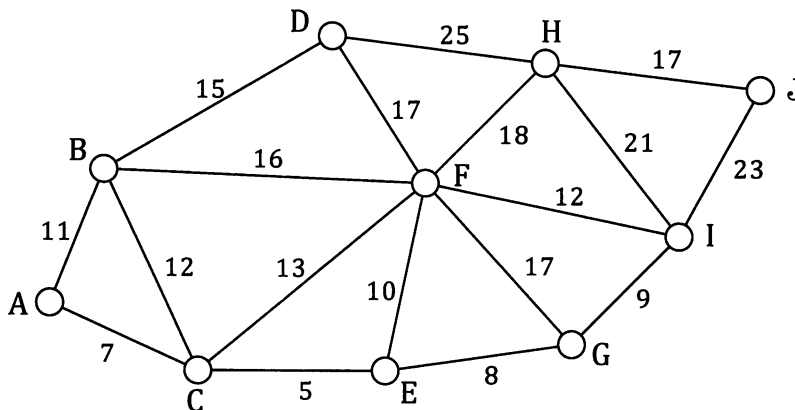
10. Five police vehicles, A, B, C, D and E have to be sent to five locations, 1, 2, 3, 4 and 5, with one police vehicle per location. The estimated time, in minutes, for each police vehicle to reach each location, from where they are now, is given in the following bipartite graph.



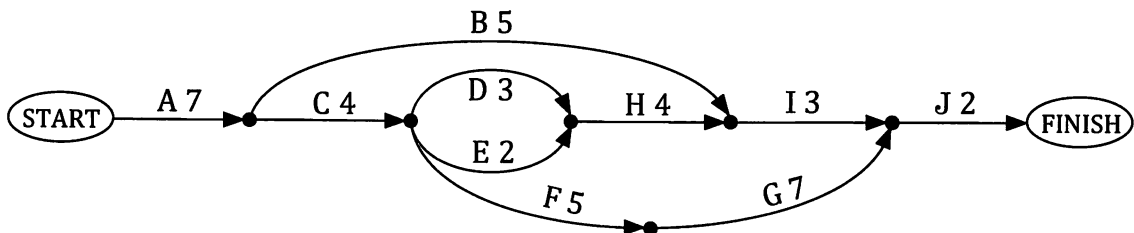
Allocate the police vehicles to the locations in such a way that the sum of the five travel times is minimised, and state the time taken for each vehicle to reach its allocated location.

11. A company expects the annual sales of one of their products next year to be 27400 and anticipate that the only significant changes in sales from one quarter to another to be those due to anticipated seasonal effects.
- If there is no quarterly seasonal effect on sales of this product roughly how many of the product should the company expect to sell each quarter?
 - In fact the sales are seasonal in nature with the seasonal indices for each of the first three quarters being as follows:
 1st Quarter 0.72 2nd Quarter 0.88 3rd Quarter 1.06
 Predict the sales numbers for this product for each quarter of next year.

12. Calculate the total *length* of the minimum spanning tree for the network shown below and show the minimum spanning tree on a diagram. (The numbers on each edge gives the *length* of that edge in units.)



13. A company identifies a number of tasks involved in the completion of a particular project. The project network below shows these tasks labelled A to J with the numbers on each arc indicating the number of days each task requires.



- Determine the minimum time to complete the project and the corresponding critical path.
- How many extra days could task E be allowed to take, over and above the 2 days already allowed for, without delaying the completion time stated in part (a)?

14. The table below shows the number of gas units used by a company every two months for a period of three years.

n	Year	Two month period	Units used
1	1	1st two months of year	1238
2		2nd two months of year	1812
3		3rd two months of year	5084
4		4th two months of year	6320
5		5th two months of year	3778
6		6th two months of year	1052
7	2	1st two months of year	1382
8		2nd two months of year	2196
9		3rd two months of year	6416
10		4th two months of year	7148
11		5th two months of year	4870
12		6th two months of year	1634
13	3	1st two months of year	1502
14		2nd two months of year	2586
15		3rd two months of year	7460
16		4th two months of year	8216
17		5th two months of year	5122
18		6th two months of year	1790

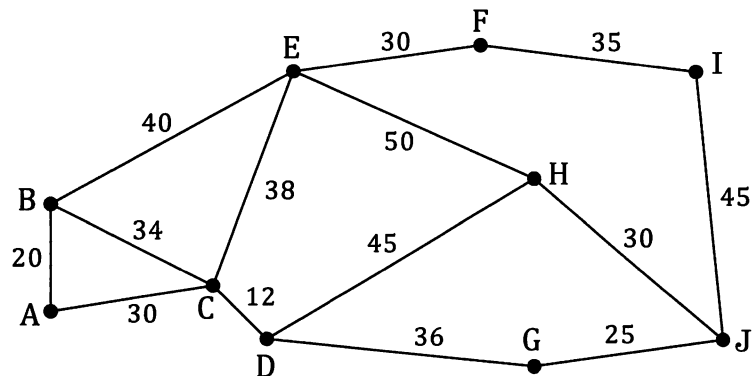
- Determine the seasonal indices for each two months of the year using the average (mean) percentage method. Give answers in decimal form and correct to four decimal places.
- Use the seasonal indices to deseasonalise the number of units used for each two month period. Give each deseasonalised figure to the nearest ten units.
- View the deseasonalised figures plotted against n to check that linear regression would be an appropriate model to use for this data pair.
- Use n and the deseasonalised data, D , to determine the least squares regression line $D = an + b$. (Give a correct to two decimal places and b to the nearest ten.)
- Use your regression line, and the seasonal indices to reseasonalise the data, to predict the number of units used for

the 1st two months of year 4.
the 4th two months of year 4.
- Suggest some events that could occur that would lead us to expect that the actual values for year 4 could differ markedly from our predicted values.

15. A company is planning the construction of a new shopping complex. The company identifies a number of tasks, the time each task will take and the order in which they must be completed:

<u>Task</u>	<u>Time</u>	<u>Order</u>
A	1 week	• Tasks B, C and D can commence provided task A has finished.
B	3 weeks	• Task E can commence provided task B has finished.
C	1 weeks	• Task F can commence provided task C has finished.
D	2 weeks	• Task H can commence provided task D has finished.
E	3 weeks	• Tasks K and I can commence provided task H has finished.
F	4 weeks	• Tasks G and M can commence provided both E and F have finished.
G	6 weeks	• Task J can commence provided both G and I have finished.
H	5 weeks	• Task L can commence provided both J and K have finished.
I	7 weeks	
J	4 weeks	
K	9 weeks	
L	2 weeks	
M	12 weeks	

- (a) Determine the minimum completion time and the critical path.
 (b) What is the maximum time that task K could take without delaying the completion of the project?
 (c) If the time for task I could be reduced to 4 weeks what would be the minimum completion time for the project?
16. A minimum spanning tree is needed for the network shown below. The number on each edge is the cost, in \$1000s, of making the connection. Find the minimum spanning tree for the network and state its total cost.



Before work is started constructing the minimum spanning tree it is found that a direct connection could be made from H to I. What would the cost of this connection need to be for it to be worthwhile including in a new minimum spanning tree?

17. One particular day three machines are to be assigned to manufacturing different products, with each machine producing a different product. There are four products required but with just the three machines working one product will not be manufactured that day. The assignment of machine to product is to be done so that the total output, in total number of units produced for the day, is maximised. Clearly showing your use of the Hungarian algorithm, allocate machines to products to achieve this, given that the numbers of units of each product that could be produced on each machine in the day is as follows:

	Product 1	Product 2	Product 3	Product 4
Machine A	2700	2300	3300	3300
Machine B	4200	3600	4100	4100
Machine C	4000	3500	3600	3300

18. A port authority wishes to build a railway network linking seven locations in the dockside complex. The network is to be constructed so that it will be possible to travel by rail from any one of the seven locations to any of the other six, perhaps not directly, but at least by going via other locations in the network. The locations are *Goods In*, *Goods Out*, *Waiting*, *Customs*, *Dock 1*, *Dock 2* and *Dock 3*. The distances involved for the feasible rail connections from any of these locations to any of the others are given in metres in the table below.

	Goods In	Goods Out	Waiting	Customs	Dock 1	Dock 2	Dock 3
Goods In	–	–	420	–	300	–	–
Goods Out	–	–	–	150	–	–	210
Waiting	420	–	–	120	240	–	–
Customs	–	150	120	–	–	210	180
Dock 1	300	–	240	–	–	195	–
Dock 2	–	–	–	210	195	–	225
Dock 3	–	210	–	180	–	225	–

- (a) Based on these distances determine the minimum length of track required for this network.
- (b) In this "minimum length network" which six pairs of locations would have a direct rail link. (Direct rail link means able to journey from one location to the other without having to pass through any of the other locations on the way.)

19. Five transport companies are asked to quote their price for four jobs involving the transportation of wide loads. The company asking for the quotes is likely to have several more such jobs in the future and wants to use a different company for each of the four jobs in order to see which company seems to give the best service. The five transport companies are *DG Transport*, *Haulage and Co*, *JA Movers*, *Move it and Shift It* and *Relocate Plus*. The prices each company quotes for the four jobs are as in the following table:



	Job 1	Job 2	Job 3	Job 4
DG Transport	\$18000	\$14600	\$28000	\$16500
Haulage and Co.	\$19800	\$12800	\$32000	\$18000
JA Movers	\$18000	\$13500	\$27000	\$17500
Move it and Shift it.	\$24000	\$12000	\$25000	\$16000
Relocate Plus	\$22000	\$13000	\$26500	\$17000

Allocate a transport company to each job with a different company for each job, one company missing out, and minimising the total cost of the four jobs according to the quotes given above. Also state what this minimum cost will be.